Scrotal Melanoma: A Systematic Review of Presentation, Treatment, and Outcomes

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BACKGROUND Primary male genital melanomas are rare. Of these, scrotal melanoma is the least common, and is unfortunately often detected late in the disease course.

OBJECTIVE To assess the clinical presentation, treatment modalities, and outcome of scrotal melanoma to aid clinician management.

METHODS Systematic literature review of PubMed, yielding 23 cases, of which 20 met the inclusion criteria.

RESULTS Although previously thought to have the best outcome of genitourinary melanomas, scrotal melanomas are associated with high mortality and late presentation. Scrotal melanoma presented as Stage I/II disease 18.75% of the time, Stage III 56.3% of the time, and Stage IV 25% of the time, whereas typical cutaneous melanoma presents as Stage I/II disease 84% of the time, Stage III 8% of the time, and Stage IV 4% of the time. Half of patients experienced recurrence of their disease. All patients who presented with metastases to distant organs died.

LIMITATIONS Small sample size. Few cases described a Breslow's depth making it difficult to draw conclusions on tumor thickness and survival outcome.

CONCLUSION Scrotal melanoma is rare, aggressive, and is often caught late in the disease course. The authors encourage dermatologists to educate patients and destigmatize genital lesions to increase the likelihood of earlier detection and better patient outcomes.

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The incidence of malignant melanoma in men is currently 2.9%.[1] Unlike most malignancies which are either stable or declining in incidence, the incidence of melanoma has been steadily increasing.[2–4] Primary male genital melanomas are an extremely rare subset, which comprise penile and scrotal melanomas. Scrotal melanoma is specifically the least prevalent of the genital melanomas, with only 23 cases reported in the literature as of December 2015. Of note, penile melanomas may be different entities from scrotal melanomas because they can involve the mucosal surface of the urethra, and management involves different surgical techniques because of the sensitive anatomical location. This review aims to assess the demographics, presenting signs and symptoms, and treatment modalities of patients with primary scrotal melanoma as there is currently no systematic review that does so. Penile melanomas are not discussed in this review.

Materials and Methods

A systematic review on PubMed/MEDLINE was performed on December 7, 2015 to gather data on patients with primary scrotal melanoma. The search terms used were "scrotal melanoma" and "scrotum melanoma." Inclusion criteria included English abstracts and articles pertaining to patients with primary scrotal melanoma. Only case reports or

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case series studies which provided individual patient information were included. Non-English articles and reply-to-author articles were excluded.

The author's search generated 67 potential articles. A total of 23 cases were identified, of which 20 met the inclusion criteria. Notably, 3 additional cases of scrotal melanoma were identified but did not meet criteria for admission into this review (one Japanese, 2 historical cases of scrotal melanoma in 1949 and 1953 that were not identified through the systematic review).[5–7]

Results

Patient Presentation

The mean age of patients with scrotal melanoma was 56.6 (range: 31–85). The most common form of presentation was a pigmented lesion. Ten percent of cases presented as cystic scrotal masses. Twenty-five percent of the patients presented with an ulcerated lesion. The average size of the lesion on presentation was 8.2 cm with a range of 1.5 to 15 cm (n = 6). Of the 11 cases that specified sidedness, 63.6% were left-sided, 27.3% were right-sided, and 9.1% were crossing the midline. Forty percent of the patients presented with inguinal lymphadenopathy. Twenty-five percent of the patients had a known mass for over 12 months before seeking medical care. Table 1 summarizes patient presentation.

TABLE 1. Scrotal Melanoma Patient Presentation

Parameter	*Result*
Age (n = 20)	
Mean (range)	56.6 yrs (31–85)
Sidedness (n = 11), %	
Left	63.6
Right	27.3
Crossing midline	9.1
Size of lesion (n = 6)	
Mean (range)	8.2 cm (1.5–15)
Breslow's Depth (n = 11)	
Mean (range)	7.3 mm (0.55–33)
AJCC clinical staging (n = 16), %	
Stage I	12.5
Stage II	6.25
Stage III	56.3
Stage IV	25

Pathological Diagnosis and Staging

Amelanotic melanoma was diagnosed in 10% of the cases. Only 11 of the cases described a Breslow's depth, with an average of 7.3 mm (range: 0.55–33 mm). The AJCC clinical staging could be determined for 80% of the patients. In Stage I 12.5% presented, 6.25% in Stage II, 56.3% in Stage III, and 25% in Stage IV. Forty-five percent of the patients had melanoma positive inguinal lymph nodes on presentation. Twenty percent of the patients had metastases to distant organs (not including lymph nodes) at presentation. All patients who presented with metastases to distant organs eventually died of their disease, with an average survival of 8 months after diagnosis (range: 1–19 months).

Treatment

The mainstay treatment in 95% of patients was wide local excision (WLE) of their lesion. One patient refused treatment and died of his disease in 6 months.[8] Twenty percent of patients had inguinal lymphadenectomies in addition to local excision. Ten percent had superficial groin dissections. Ten percent had orchiectomies in addition to local excision. Twenty-five percent of patients were concomitantly treated with chemotherapy in addition to WLE. All these patients either had positive lymph nodes or metastases to distant organs. Of note, Mohs surgery has recently been investigated for the treatment of melanoma in situ and melanoma. However, further research with this modality is needed before Mohs surgery can be confidently recommended as the standard of care for scrotal melanoma.

A variety and different combination of chemotherapeutic agents were used for patients, including dacarbazine, cisplatin, visblastine, IL-2, interferon, vincristine, doxorubicin, 5-Fluorouracil, tumor-derived vaccine, vindesine, and temozolomide. The most common chemotherapeutic agent used was dacarbazine. Three patients with metastatic disease were additionally being treated with radiation.[9–11]

Outcomes

Sixty percent of patients died of their disease; 5% died of complications of chemotherapy; 5% died with no

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evidence of disease. Twenty-five percent of patients were alive with no evidence of disease; 5% had no follow-up. These results are summarized in Table 2. Of the 12 patients who died of their disease, the average length of time from diagnosis to death was 58.9 months (range: 1–461 months). If the individual who lived 461 months[9] after diagnosis is considered an outlier, and not included in the analysis, then the average length from diagnosis to death would be 32.1 months (range: 1–80 months). Of the 5 patients who had no evidence of disease when their cases were reported, only 2 of them were followed for more than 5 years.[9,12] The patient who survived longest after diagnosis was alive with no evidence of disease 17 years after diagnosis, although he had 2 incidents of recurrence at 32 and 60 months.[13] Table 3 presents specific patient treatment and outcome.

Half of patients eventually had recurrence of their disease after initial treatment. Sites of metastasis and recurrence include the pelvis, brain, skeletal system, liver, lungs, abdominal wall, and skin. Ninety percent of recurrences occurred within 3 years (average = 17.5 months), and one case seemed to be an outlier with recurrence occurring at 432 months after diagnosis. The average time from recurrence to death was 22 months (range: 0–56 months). Eighty percent of patients who had recurrence were treated with chemotherapy. Only 1 patient had genetic tumor analysis completed and was reported to be KIT negative.[16]

Discussion

Scrotal melanoma is recognized as the rarest form of genitourinary melanoma. Due to the paucity of data on scrotal melanoma in the literature, there are no current guidelines to aid clinicians in its diagnosis or management.

TABLE 2. Scrotal Melanoma Survival Outcome

Survival Outcome (n = 20)	*Result, %*
Died of disease	60
Alive, no evidence of disease	25
Died of chemotherapy complications	5
Died, no evidence of disease	5
No follow-up	5

The most common form of presentation for a scrotal melanoma was a raised pigmented lesion; however, it can also present as an ulcerated lesion, a cystic mass, or a mass with satellite lesions. Three of 20 cases were initially misdiagnosed as either an epidermal inclusion cyst, benign scrotal cyst, or a blood blister.[12,19] Therefore, a thorough and broad differential diagnosis should always be constructed for scrotal lesions, with a consideration for both benign and malignant etiologies. Examples include melanocytic nevi, bowenoid papulosis, Bowen's disease, extramammary Paget's disease, squamous cell carcinomas, and pigmented basal cell carcinomas.[17] For amelanotic and cystic lesions, the differential diagnosis becomes even more comprehensive to include many diseases of mesenchymal origin. Liposarcomas, leimyosarcomas, rhabdomyosarcomas, malignant fibrous histiocytomas, epidermal inclusion cysts, mesotheliomas, and lymphomas are among those that should be included.[20]

Once the diagnosis of scrotal melanoma has been made, treatment options should be offered to patients. First, a thorough oculocutaneous examination should be performed to ensure that the scrotal melanoma is in fact the primary malignancy. Next, a Breslow's depth must be established to aid in determining excision margins and staging of the melanoma. Unfortunately, one limitation in this study is that only 55% of the cases in this review reported a Breslow's depth, making it difficult to make concrete conclusions on tumor thickness and clinical outcome.

The gold standard treatment of scrotal melanoma and the therapy offered to 100% of patients in this review is WLE of the lesion. No patient in this review had local recurrence of their disease. For local excision, the authors believe that the National Comprehensive Cancer Network (NCCN) offers sufficient guidelines to determine margins based on tumor thickness.[21] Furthermore, it is important to assess neighboring structures within the scrotum and clinical judgment should be used to determine whether any additional structures should be removed. For example, 2 of the patients in this review had concomitant orchiectomies.

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TABLE 3. Specific Patient Treatment and Outcome

References	*Age*	*Treatment*	*Follow-up (Time)*
Agrawal and Kumar[14]	45	WLE, right inguinal lymphadenectomy	DOD (1 mo)
Das Gupta and Gupta[15]	65	WLE, chemotherapy (dacarbazine)	n/a
Lillis and colleagues[16]	62	WLE	DOD (6 months)
Vasudeva and colleagues[8]	65	Refused treatment	DOD (6 months)
Sánchez-Ortiz and colleagues[9]	31	WLE	DOD (461 months)
Sánchez-Ortiz and colleagues[9]	65	WLE, bilateral inguinal lymphadenectomy	Died, with NED
Sánchez-Ortiz and colleagues[9]	39	WLE, chemotherapy (vincristine, bacillus calmette-guerin, dacarbazine)	DOD (47 months)
Sánchez-Ortiz and colleagues[9]	59	WLE	DOD (26 months)
Sánchez-Ortiz and colleagues[9]	64	WLE	DOD (31 months)
Sánchez-Ortiz and colleagues[9]	61	WLE, right inguinal lymphadenectomy, chemotherapy	Died of chemotherapy complications, with NED
Damala and colleagues[17]	55	WLE by scrotectomy, chemotherapy (temozolomide)	DOD (1 mo)
Nguyen and colleagues[12]	50	WLE, sentinel lymph node biopsy, superficial groin dissection	NED (34 mo)
Nguyen and colleagues[12]	64	WLE	NED (80 mo)
Berkmen and colleagues[18]	69	left hemiscrotectomy, chemotherapy (decarbazine, vindesine, cisplatin)	DOD (6 mo)
Berkmen and colleagues[18]	38	left hemiscrotectomy and high orchiectomy	NED (50 mo)
Konstadoulakis and colleagues[13]	55	WLE	NED (17 yrs)
Konstadoulakis and colleagues[13]	48	WLE	DOD (68 mo)
Moul and colleagues[19]	85	Hemiscrotectomy, left inguinal orchiectomy, left inguinal lymphadenectomy	DOD (19 mo)
Davis and colleagues[10]	61	left hemiscrotectomy	DOD (48 mo)
Ray and colleagues[11]	50	WLE, left superficial groin dissection	NED (69 mo)

DOD, died of disease; NED, no evidence of disease; WLE, wide local excision.

Next, the physician must assess lymph nodes. According to the NCCN Practice Guidelines for Oncology, physicians should consider a sentinel lymph node biopsy for Stage IA melanoma with adverse features (e.g. lymphovascular invasion, tumor-infiltrating lymphocytes) and encourage sentinel lymph node biopsy for Stage IB or II.[21] Patient's with Stage III disease should be offered lymph node dissection according to guidelines. Although all patients in this review who underwent local excision achieved effective control from local recurrence, the same cannot be said about regional and distant recurrence. Of the 10 patients who presented with no clinical adenopathy and no distant metastases who did not receive inguinal node staging, 70% had either regional recurrence or distant metastases. The majority, 71% (5/7) of those cases included regional recurrence.

It was previously thought that scrotal melanomas may be the least aggressive and have the best outcome out of all genitourinary melanomas.[12] However, this was based on a case series with only 2 patients who were alive and with no evidence of disease after excision. Although the limited sample size in this review makes it difficult to draw strong conclusions on the aggressiveness of scrotal melanomas, their tendency to present late in the disease course is more conclusive. Typical cutaneous melanoma patients present with localized cutaneous disease (Stage I or II) in 84% of cases,[22] whereas just 18.75% (n = 16) of the patients with scrotal melanoma presented with Stage I or II disease. Likewise, only 8% of typical melanoma patients present with Stage III disease, compared with an alarming 56.3% of scrotal patients. Last, just 4% of typical cutaneous melanoma patients presented with Stage IV disease

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(distant metastases), whereas 25% of scrotal melanoma patients presented with Stage 4 disease, further providing evidence of their late presentation. Table 4 presents information on the patients who presented with metastatic disease to distant organs. Because very few patients in this review were shown to survive greater than 5 years it is difficult to compare and assess survival outcomes of scrotal melanoma to typical cutaneous melanoma. Because more cases are reported in the literature, it may become more clear if the poor outcome of scrotal melanomas can be attributed to a more aggressive nature or solely to a later presentation.

Another issue worthy of discussion is both the patient's and clinician's comfort in dealing with genital lesions. Twenty-five percent of the patients in this review had a known mass for over 12 months before seeking medical care. Equally disturbing, 3 patients in this review had a mass that was 10 cm or greater. A study on male genital lesions showed that 65% of patients had a delay of 6 months before consulting a medical professional, citing embarrassment because of symptoms in a sexualized area as the primary reason.[23] In addition, 50% of the patients in that study did not even discuss the problem with their partners. It has also been shown that men have less knowledge of cancer symptoms and more passive detection behavior than women.[24] Therefore, the authors believe that it is vital for both dermatologists, urologists, and primary care physicians to routinely educate and discuss genital malignancies with male patients in effort to shorten the delay of diagnosis and to destigmatize the subject. A study on full-body skin examinations (FBSEs) highlights this point by showing that patient education on the objectives and benefits of a FBSE decreases patient concern about receiving a genital examination.[25] Furthermore, it was shown that only 42% of patients reported examining themselves for skin cancer.[26] The authors hope that with education and destigmatization of genital lesions, patients will be encouraged to seek early care for concerning scrotal lesions, thus increasing the likelihood of having improved outcomes because of less advanced disease on presentation.

TABLE 4. Patients With Metastatic Disease on Presentation

Patient	*Metastatic Site*	*Outcome*
1[16]	Skeletal, hepatic	DOD (6 months)
2[17]	Skeletal, hepatic, adrenal	DOD (1 mo)
3[18]	Hepatic, pulmonary	DOD (6 months)
4[19]	Pelvic	DOD (19 months)

DOD, died of disease.

As scrotal melanoma is an extremely rare entity, this posed challenges and limitations to this study. First, this study is limited with respect to a small sample size. There were only 23 reported cases, of which 20 cases met criteria for inclusion. Another limitation is that all the data were derived from case reports, putting the literature at risk for potential bias. However, as this disease is so rare, there is currently no other means to gather information on scrotal melanoma. Therefore, although these limitations do in fact exist, the authors still consider it worthwhile to gather and analyze the data of these case reports in aggregate as this is currently the only way to learn more about this rare disease.

Conclusion

In conclusion, although scrotal melanomas are rare, their late presentation and subsequent poor outcome makes it imperative that dermatologists educate their patients and destigmatize genital lesions to increase the likelihood of earlier detection and better patient outcomes.

References

1. Siegel R, Ma J, Zou Z, Jemal A. Cancer statistics, 2014. CA Cancer J Clin 2014;64:9–29.
2. Simard EP, Ward EM, Siegel R, Jemal A. Cancers with increasing incidence trends in the United States: 1999 through 2008. CA Cancer J Clin 2012;62:118–28.
3. Jemal A, Saraiya M, Patel P, Cherala SS, et al. Recent trends in cutaneous melanoma incidence and death rates in the United States, 1992-2006. J Am Acad Dermatol 2011;65:17–25.
4. Shaikh WR, Dusza SW, Weinstock MA, Oliveria SA, et al. Melanoma thickness and survival trends in the United States, 1989 to 2009. J Natl Cancer Inst 2015:108:1–7.
5. Sasaki H, Ishihara K. Clinical symptoms and prognosis in malignant melanoma of the pubic region [in Japanese]. Gan To Kagaku Ryoho 1989;16:1721–7.
6. Higgins CC, Warden JG. Cancer of the scrotum. J Urol 1949;62:250.

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7. Allen AC, Spitz S. Malignant melanoma: a clinicopathological analysis of the criteria for diagnosis and prognosis. Cancer 1953;6:1.

8. Vasudeva P, Agrawal D, Goel A. Malignant melanoma of the scrotum. Urology 2008;71:1053–4.

9. Sánchez-Ortiz R, Huang SF, Tamboli P, Prieto VG, et al. Melanoma of the penis, scrotum and male urethra: a 40-year single institution experience. J Urol 2005;173:1958–65.

10. Davis NS, Kim CA, Dever DP. Primary malignant melanoma of the scrotum: case report and literature review. J Urol 1991;145:1056–7.

11. Ray B, Huvos AG, Whitmore WF Jr. Unusual malignant tumors of the scrotum: review of 5 cases. J Urol 1972;108:760–6.

12. Nguyen AT, Kavolius JP, Russo P, Grimaldi G, et al. Primary genitourinary melanoma. Urology 2001;57:633–8.

13. Konstadoulakis MM, Ricaniadis N, Karakousis CP. Malignant melanoma of the scrotum: report of 2 cases. J Urol 1994;15:161–2.

14. Agrawal R, Kumar MJ. A malignant transformation in a pre-existing benign scrotal naevus. Clin Diagn Res 2012;6:1562–3.

15. Das Gupta HK, Gupta R. Malignant melanoma of scrotal skin. Indian J Dermatol 2011;56:223.

16. Lillis JV, North J, Vetto JT, Corless CL, et al. Primary scrotal melanoma presenting as a large, amelanotic, exophytic mass. Arch Dermatol 2009;145:1071–2.

17. Damala K, Tsanou E, Pappa L, Sintou-Mantela E, et al. A rare case of primary malignant melanoma of the scrotum diagnosed by fine-needle aspiration. Diagn Cytopathol 2004;31:413–6.

18. Berkmen F, Tandoğdu R, Ardjçoğlu AJ. Primary scrotal malignant melanoma. Report of 2 cases and review of the literature. Exp Clin Cancer Res 1998;17:91–3.

19. Moul JW, Ho CK, McLeod DG. Primary malignant melanoma of the scrotum. Int Urol Nephrol 1992;24:641–3.

20. Woodward PJ, Schwab CM, Sesterhenn IA. From the archives of the AFIP: extratesticular scrotal masses: radiologic-pathologic correlation. Radiographics 2003;23:215–40.

21. *National Comprehensive Cancer Network. Melanoma (Version 2.2016).* Available at: http://www.nccn.org/professionals/physician_gls/pdf/melanoma.pdf. Accessed December 24, 2015.

22. *National Cancer Institute SEER stat fact sheets: Melanoma of the skin.* Available at: http://seer.cancer.gov/statfacts/html/melan.html. December 24, 2015.

23. Skeppner E, Andersson SO, Johansson JE, Windahl T. Initial symptoms and delay in patients with penile carcinoma. Scand J Urol Nephrol 2012;46:319–25.

24. de Nooijer J, Lechner L, de Vries H. Early detection of cancer: knowledge and behavior among Dutch adults. Cancer Detect Prev 2002;26:362–9.

25. Leffell DJ, Berwick M, Bolognia J. The effect of pre-education on patient compliance with full-body examination in a public skin cancer screening. J Dermatol Surg Oncol 1993;19:660–3.

26. Federman DG, Kravetz JD, Tobin DG, Ma F, et al. Full-body skin examinations: the patient's perspective. Arch Dermatol 2004;140:530–4.

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